

CReBot: Exploring Interactive Question Prompts for Critical Paper Reading

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Abstract

Pre-compiled guidelines with a static question list can stimulate critical thinking during scientific paper reading. However, they could be less engaging than taking live question prompts from others. In this paper, we develop CReBot that interactively asks section-level critical thinking questions and customize it for routine paper readers with prior research experience and novices new to research. Our first within-subjects study with 24 routine readers demonstrates CReBot's engagement and usefulness over static guidelines. Then, with more teacher-like question-specific hints prepared for CReBot, we conduct another within-subjects study with 20 novices. The results, however, indicate that CReBot might not be better than static guidelines for beginners. Nevertheless, both user groups favor CReBot's contextualized questions and interaction flexibility. We conclude by design implications for interactive tools to facilitate critical reading.

Keywords: Paper reading, interactive support, bot, question prompts critical thinking

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1. Introduction

Critical thinking is one of the higher-order skills that university students need to learn [1, 2]. When applied to reading scientific publications, it requires readers to question the texts as they study a claim, a method, a result, or other contents in a paper [3]. For example, a critical reader should ask whether the evidence convincingly supports a paper's claims and why these claims matter to the audience [3]. Critical reading guidelines or prompts that consist of a checklist of questions provided by experienced researchers are great materials that encourage critical thinking ([4, 5, 6, 7]), e.g., "What are the authors trying to do in writing this? How convincing is what they are saying? And in conclusion, what use can I make of this?" [3].

While these checklist-like question prompts serve as a good starting point for practicing critical paper reading, they are rather static and thus tend to be less engaging and useful than receiving interactive questions and opinions about the paper from others (e.g., peers, teachers, supervisors) [8, 9]. For example, in a group meeting, colleagues can usually raise critical thinking questions about the presented paper content, stimulating the presenter to think of the paper from different aspects. For paper reading on computer screens, existing works have explored possible means to facilitate critical thinking, such as online collaborative reading platforms [10] and projection of teachers' reading patterns on digital papers [11]. However, these methods require human peers or teachers to read the same materials beforehand or at the same time to raise suitable questions. They hence fall short in the scalability as such qualified social questioners are not always available.

A bot is a possible alternative to support critical paper reading by imitating how humans interactively ask questions. Previous works have demonstrated the strengths of bots that can interactively question the learners in the educational domain [12, 13], e.g., learning factual knowledge [12] or programming concepts [13]. Nevertheless, little work has looked into the design, usefulness, and user experience of an interactive question-prompting bot for critical paper reading. Unlike those knowledge acquisition tasks in which there is usually a correct answer for each question, an interactive critical thinking facilitator should encourage readers to develop their own understanding and interpretation [3, 14]. Also, it should offer users sufficient critical thinking support while mitigating the possible interference in the paper reading process [15]. Furthermore, the design and usefulness of such interactive facilitation may vary across the users' levels of reading and research

experience with the paper topics, as previous research suggests that novices generally desire more assistance and guidance and perform differently in their tasks compared with seniors [16]. Therefore, despite the success of interactive bots in other educational scenarios, gaps exist in exploring the design and usage of a question-asking bot to engage users with different levels of research experiences in critical paper reading.

To this end, we develop CReBot (**Critical Reading Bot**) that asks section-level questions and provides guidance in real-time to facilitate users in critical paper reading and explore its usage in reading HCI (Human-Computer Interaction) publications. Specifically, our design and evaluation of CReBot are customized for two types of potential users in two stages. In the first stage, we target the CReBot as a peer-like agent for assisting routine paper readers who generally have prior experience in research and paper reading, e.g., postgraduate students. As they read each section of a paper in a web browser, CReBot, located in the sidebar, prompts associated questions from the critical paper reading guidelines pre-compiled by our design team. Users can respond to the bot's question, switch to another question of the same or a different section, and add their own critical thoughts any time they want. We evaluate this CReBot against the conventional checklist of instructions and questions (denoted as "Guideline") via a mixed-methods, within-subjects study with 24 routine paper readers, including 19 graduates and five undergraduates with prior research experience. The results show that CReBot better engages these routine paper readers in the reading process by encouraging more critical reading behaviors, and it is perceived significantly more useful and easier to use. Participants suggest that such interactive facilitation from CReBot might also be useful for novices who are new to scientific research or seldom read academic papers before.

Inspired by the first-stage findings, we refine CReBot as a teacher-like agent specifically for novices in the second stage. First, we verify the novices' needs and identify their requirements of CReBot via a Speed Dating study [17, 18], where we present eight novices with storyboards of CReBot usage scenarios and with low-fidelity prototypes for user enactments. Then, based on the Speed Dating results, we mainly augment the previous CReBot design by offering hints from experts that help novices look for possible answers to each prompted critical thinking question. Next, we evaluate the refined CReBot for novices via a within-subjects study with 20 undergraduate students who have little or no research experience. Experimental results show that CReBot achieves comparable performance as the Guideline in engaging

novices in the critical reading process and helping them comprehend and criticize the paper content. Both tools are deemed equally useful and easy to use by these participants. Novices generally favor CReBot's interactive questions and hints for critical paper reading but feel that the CReBot's interaction sometimes cause interruptions as it does not match their traditional reading habit.

In sum, we contribute a bot prototype to explore interactive question prompts for critical paper reading support. Our two-stages investigation add to the understandings of how people with different levels of paper reading experience perceive and read scientific publications with such a bot. We also discuss the generality of CReBot and propose the design considerations for improving the usefulness and user experience of interactive technologies for paper reading and critical thinking support. Meanwhile, we contribute a question bank that encourages critical thinking of research publications.

2. Related Work

2.1. Critical Paper Reading and Its Guidelines

The ability to think critically is widely regarded as an essential skill people should gain through high-level education [1, 2]. Critical thinking requires not only understanding the problem, text, or answer (i.e., *comprehension*) but also “making clear, reasoned judgments” (*criticism*) to them [19]. For those who need to read scientific papers to learn from existing literature and get insights for their works, it is crucial to think critically in their reading practices [3]. To do that, readers should actively listen to what a text says and respond in their own voice, e.g., they ask questions, think of examples to challenge the text, and relate it to their purposes or experiences [8, 20]. To learn and exercise critical reading skills, people can get guidance from senior scholars, who usually give tutorials and organize classroom reading activities [8, 21]. In a more accessible way, they can also refer to online critical thinking questions and guidance compiled by experienced researchers [3, 6, 4, 14, 5]. For example, to read critically, a handout from the University of Toronto recommends that people should analyze “what the patterns of the text are”, interpret “what the patterns of the argument mean”, and evaluate “how well the text does what it does and what its value is” [4]. Such guidelines could be useful for both novices and those who already have some experience in critical paper reading, as a written set of general rules, principles, or pieces

of advice from others are generally helpful for users to complete given tasks [22, 23, 24, 25, 26].

While these critical reading guidelines (denoted as “Guideline” condition in the experiments) traditionally contain a checklist of question prompts that users can ask about the text, they are static and generic, which could be less engaging and useful than the interactive questions from others. As argued by Lev Vygotsky in the theory of scaffolding [27], we learn best when we socially interact with others, who offer us questions and hints that are adaptive to our status and help us improve. Ngoon et al. provided empirical evidence to this insight by showing that the interactive techniques – adaptively presented suggestions and guidance – improve the quality of feedback from novice reviewers on creative work, compared to the static ones [28]. In this paper, we explore the design of interactive tools for reading papers critically, and we compare our tool against the Guideline to evaluate its usefulness and user experience with both routine paper readers and novices.

2.2. Interactive Tools to Facilitate Paper Reading

Researchers in HCI (Human-Computer Interaction) have explored various interactive tools to facilitate people’s reading experience. These tools can be categorized into two groups regarding how they manipulate the reading materials. The first group of approaches does a content “reduction” usually for helping people skim a text and grab its main idea [29, 30, 31, 32]. For example, Kobayashi et al. proposed to sequentially fade-out characters sentence-by-sentence for each paragraph and demonstrated its effectiveness in improving readers’ text comprehension [30]. Kim et al. presented an interactive reading system that automatically links document text with corresponding table cells and showed that it increased users’ accuracy and speed in navigating the matched sentences of table cells [31]. The other way of reading assistance “enriches” the reading materials with extra resources and guidance [33, 34, 35, 36, 37, 38]. For example, Khan et al. designed an audio skimming app that blends auditory and visual reading for situational impairments (e.g., walking) [34]. Head et al. introduced an augmented reading interface that can provide users with easy access to the definitions of technical terminology and mathematical symbols [15]. To help users quickly track and glance related work, Wang et al. developed a visualization system that presents the literature review as interactive slides [33].

Existing ideas of interactive facilitation for paper reading fall into the content enrichment type of methods [39, 11]. For example, Cheng et al.

developed a SocialReading system that shares teachers' gaze data for an academic paper and validated that it can improve students' reading comprehension of that paper [11]. Tan et al. designed and implemented WiREAD, which offers a collaborative environment with practices, questions, and discussions for peers and instructor to engage in critical paper reading together [10]. Unlike the tools mentioned above, CReBot can facilitate critical reading via question prompts without the need to have teachers or peers read the same materials beforehand or at the same time.

2.3. Interactive Bots in Educational Domains

According to the Google's English dictionary provided by Oxford Languages, a 'bot' is an autonomous program on the internet or another network that can interact with systems or users. The bot's accessibility and ability to imitate human behaviors make it increasingly popular in educational domain [40, 41]. For example, the tutoring system AutoTutor has been used to teach college students in multiple domains, such as computer literacy and critical thinking [42]. AutoTutor provides explanations, feedback, scaffolding, deep reasoning questions, and subject content in online courses, and multiple studies have demonstrated its effectiveness in improving learning gains [42]. Similarly, Wambsganss et al. designed ArgueTutor that judges argumentative writing performance of users' essay and suggests how to improve [43]. Educational bots commonly use questions as a start to engage users in learning. For example, Winkler et al. developed Sara that acts like a teacher to asks students questions during an online video lecture [13]. They demonstrated in a lab experiment that Sara could significantly improve learning than the without-Sara condition in a programming learning task [13]. Ruan et al. created QuizBot, an interactive agent that asks questions and provides feedback to users' answers in learning factual knowledge about science, safety, and English vocabulary [12]. They showed that QuizBot engaged users better in the learning process than the traditional flashcard tool, and users preferred the bot strongly for casual learning [12].

Inspired by these successful educational bots, we explore the possibility of an interactive bot for critical paper reading support. Our CReBot uses questions as the main interactive materials because they are generally effective in encouraging thinking [44, 45, 46]. CReBot does not give corrective feedback to readers' responses as Sara and QuizBot do in their knowledge acquisition tasks with standard answers. Instead, it focuses on prompting suitable critical thinking questions and encouraging users with hints to find

or think of the answers [3, 4]. In all, to the best of our knowledge, our work is the first to probe the design, usefulness, and user experience of interactive question prompts from a bot for critical paper reading.

3. Stage 1: Developing CReBot for Routine Paper Readers

Our exploration consists of two stages. In the first stage, we target CReBot as a peer-like assistant those who generally have paper reading experience and need to read scientific papers very often. Critical paper reading is an important requirement for them [3], and CReBot that imitates the colleagues' question-asking behaviors could be a good facilitator in this process. In this paper, we treat them as “**routine paper readers**” to differentiate this user group from Stage 2's **novices** who have little paper reading experience or are new to research. These **routine paper readers** can be full-time researchers in companies, postgraduate students, or undergraduates that have deep engagement in paper reading and research. In this section, we first present how we compile the critical reading guidelines and customize them in the HCI paper domain as a case demonstration. We then describe our design and implementation of CReBot, a proof-of-concept prototype that facilitates users to read paper critically by interactively prompting section-level questions from the guidelines on the fly. Next, we report an empirical evaluation of CReBot with our Stage 1 target users and present the results. We aim to explore the following research questions:

RQ1: Compared to the conventional Guideline that outlines the critical reading instructions and questions, how would CReBot affect the *routine paper readers*' behaviors, perceived engagement and difficulty in the process, and perceived performance in the outcome of critical paper reading?

RQ2: How would the *routine paper readers* use the CReBot and Guideline in practice, and what is their acceptance towards such tools for critical paper reading?

3.1. Critical Reading Guidelines with Questions Customized in HCI Domain

To begin with, we compile critical paper reading guidelines based on the related and publicly available articles, tutorials, and books [4, 3, 5, 47, 6, 48]. The guidelines start with 0) *setting a reading goal* that can help readers establish a constructive purpose for reading the paper critically [3] and 1) *understanding the general idea of the paper* from the title, abstract, introduction, heading of each section, and conclusions [3, 6]. When 2) *digging*

into each section, a critical reader should first *comprehend* its content (i.e., identify **what** the authors are trying to do) and then *criticize* it by examining **how** the authors achieve their purposes, understanding **why** the authors do it in this way, and assessing **how well** the authors' writing and logic are [3, 4, 44]. Finally, readers can reflect on the paper by summarizing its strengths, weaknesses, and how it contributes to their reading goals [3, 4, 47].

Next, we customize the prototypical questions of guidelines into the HCI paper domain as a demonstrated case of CReBot, since the structure of a paper and the contextual critical thinking questions for each paper section could vary across different research fields. Using papers published on CHI2019, a top HCI venue [49], as an example, we showcase our **three-phases question customization process** as below:

1. Identify common paper sections. We first collect all CHI2019 proceeding papers and late-breaking works (LBW) from its website and extract their outlines using the PyPDF2 package [50], which captures the papers' section and subsection titles. We sort these titles by their numbers of occurrences and include the frequently used ones for further analysis (in our case: > 3 for proceeding papers, 168 items; > 1 for LBW, 93 items). Then, two authors independently go through these headings and inductively group them into potential categories. After determining the categories of common (sub)sections through two rounds of comparisons and discussions, they assign the headings into each category independently. The intraclass correlation coefficient (ICC), a standard measure for quantifying the degree to which a fixed number of raters have consistent judgments [51], is 0.761, suggesting a good consistency in the coding [52]. Finally, they resolve the disagreement via discussion and **output 19 categories of sections with commonly used titles (Fig. 1).**

2. Collect section-level questions via interviews with domain researchers. To collect a question bank for critical reading in the HCI domain, we conduct semi-structured online interviews with four HCI researchers (three males, one female; age: $M = 28.8, SD = 1.7$) using a video conference tool Zoom [53]. They all have over three years of experience in reading HCI papers (90+ papers per year), submitting CHI papers (at least one accepted), and reviewing CHI papers. **[Procedure]** In the interview with each participant, we first present the general guidelines and ask for their relevance to personal critical reading experiences. Then, we invite the interviewee to recall what critical thinking questions are in mind when reading paper (sub)sections in each of the 19 categories, with example titles, contents,

and prototypical questions on the shared materials ¹. In the end, we ask for their research experience and common means to exercise critical paper reading. We compensate each participant \$10 for about 50 minutes spent in the study. **[Results]** All interviewees agree that the general guidelines are consistent with their critical reading experiences. They usually read HCI papers about creativity support, virtual reality, gamification, education, visualization, human-AI interaction, and voice interface. They mention three ways for learning critical paper reading: experience of and instructions received for reviewing HCI papers; reviewers' comments on their own submissions; and questions from lab mates or advisors. Therefore, apart from the 276 section-level questions from the interviewees, we also add two other sources to enrich the question bank: 1) guidelines for reviewers in CHI papers (e.g., [54]); and 2) audiences' questions about papers shared in weekly HCI group meetings for three weeks at a local university. In total, we collect **363 critical thinking questions for common (sub)sections in CHI publications**.

3. Assign critical thinking aspects and levels to each question. As mentioned by our interviewees, readers can think critically about a (sub)section from different aspects, e.g., the *motivation* and *results* in the Introduction. Inspired by this, two authors who majored in HCI further contextualize the questions by first identifying the aspects in each question category via two rounds of coding and discussions on the 363 questions. They then independently label the questions in each aspect into levels of *what*, *how*, *why* or *how well*. They reach an excellent level of agreement over the labels (ICC = 0.920) and discuss and resolve the conflicts. Fig. 1 shows the example **section categories and corresponding questions with critical thinking aspects and levels** ².

3.2. CReBot System Design and Implementation

3.2.1. Design Criteria.

When facing design choices of CReBot that interactively prompts questions to readers, our decisions are guided by three criteria. **C1)** Reproducibility: We adopt publicly available technologies so that other researchers and practitioners can easily reproduce the system and generalize it to other paper domains [55]. **C2)** Practicality: We prefer technologies that do not require

¹The interview materials are attached in the supplementary materials.

²The full set of CHI section categories and customized critical thinking questions is attached in the supplementary materials.

Category	Aspect	Example question	Level	Category	Aspect	Example question	Level
Abstract	• Background: What is the general background of this paper? (1)			Participant	• Sample size: Is the sample size of the participants enough? (4)		
	• Target: Who is it in dialogue with? (1)				• Background / Demographics: What is the background of the participants? (1)		
	• Proposed work: How did the authors solve the problems? (2)				• Recruitment: Are the participants required to have domain knowledge? (2)		
	• Findings: What is the conclusion of the paper (1)				• Validity: Are the participants representative as targeted users? (4)		
	• Relevance: Is this paper related to your research interests? (4)			Task	• Purpose: How do the authors decide this task? (2)		
Introduction	• Background / Problem: What is the background of the problem? (1)				• Design: Is this task well-designed? (4)		
	• Motivation: Why should we care about this research problem? (3)				• Clarity / Replicability: Do the authors clearly explain the task? (4)		
	• Proposed work: Is the method of this work novel? (4)			Procedure	• Details: Why are the authors running the procedure in this way? (2)		
	• Findings: What is the benefit of the findings (4)				• Payment: Do the participants get reasonable payment for their time? (4)		
	• Contribution: Why do the contribution and benefit matter? (3)				• Clarity: Does the procedure include any necessary detail? (4)		
Related Work	• Logic / Structure: What categories of related works are presented? (1)			Measures	• Purpose: What is the purpose of each measurement? (1)		
	• Previous work: Why are people doing this research? (3)				• Details: Do the measures reflect the variables properly? (4)		
	• Need of this work: How is this problem situated in the literature? (3)				• Validity: Have these measures previously used in literature or not? (2)		

Figure 1: Example categories of sections/subsections in CHI papers, corresponding questions with critical thinking aspects and levels (1 - what, 2 - how, 3 - why, 4 - how well).

installation of extra hardware/software and can be easily accessed by users [56, 55]. **C3)** Familiarity: We favor an intuitive interface design that is easy to learn [57].

3.2.2. User Interaction with CReBot.

We build CReBot as a responsive web-based application (Fig. 2a) embedded in a paper reading web page powered by react-pdf-highlighter [58] and pdf-js [59] (C1, C2). The right part of the interface (i.e., part *i* in Fig. 2) displays the paper content which users can zoom in/out, highlight, and comment on (C3). CReBot stays in the left sidebar to set up users' expectation of where to interact with (C3). Following the interface design of educational bots [13, 12, 55], CReBot displays the interaction history (*ii*) and its current message (*iii*) in a chat window (C3). Users can receive from CReBot the critical thinking questions with context indicated by "category $\Rightarrow$ aspect $\Rightarrow$ level" on top and the general guidance for this level of questions by hovering the mouse on "Tip". To interact with CReBot, users can 1) type down and "send" their critical thoughts in reply to the current question; 2) see "next question" of the same section; 3) "add their critical questions" to the section if any; and 4) switch to questions of a specific section or subsection (*iv*). When ready, users can click "Finish reading" to get questions and guidance to reflect on the paper and "export all chat log" to record their critical reading notes with CReBot (*v*).

3.2.3. Backend Server and Interaction Logic

After the web app loads in a paper, it extracts the paper's outline and sends it to a python flask server, which uses the fuzzywuzzy package [60] to compute similarity scores between each (sub)section title and our collection of heading items in the 19 categories (e.g., item "Design implication" in category

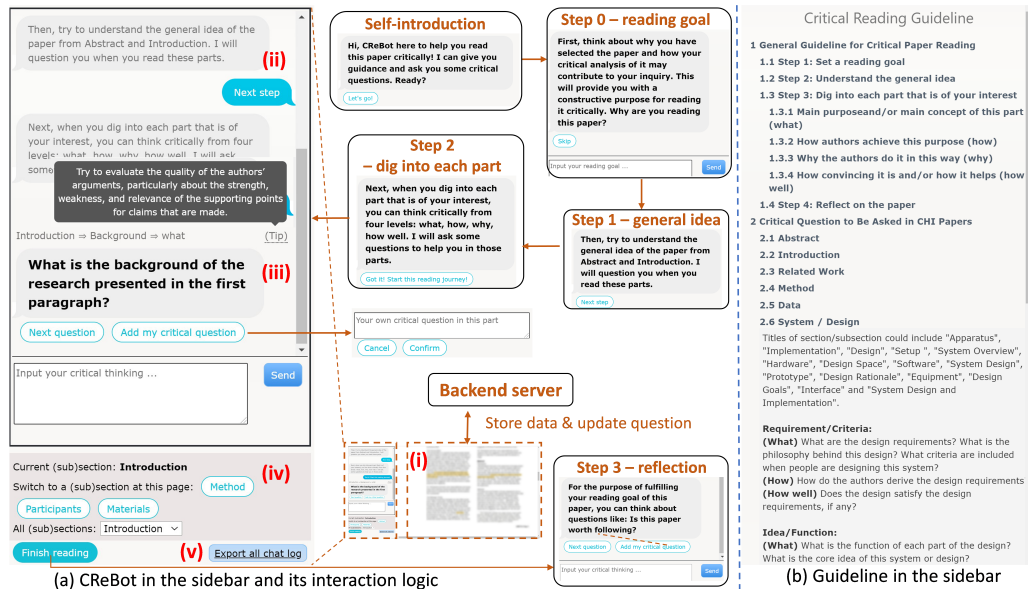


Figure 2: (a) CReBot and interaction logic following the compiled critical paper reading guidelines: i) Interface for the reading material in both CReBot and Guideline conditions; ii) Interaction history; iii) Current message; iv) A panel for switching a (sub)section; v) Buttons to proceed and export the reading notes. (b) The Guideline tool that documents the critical paper reading guidelines.

“Implication”). It returns the (sub)sections of the loaded paper that have similarity scores over a threshold (90% in our case) as the candidates for CReBot to ask questions about (Fig. 2a part *iv*). After a self-introduction, CReBot follows the steps of guidelines by first asking users for reading goals and encouraging them to grasp the paper’s general idea. It then stimulates the readers to dig into each section of the paper and prompt relevant critical questions. CReBot will prompt a new question every time readers answer the current one or click “Next Question” using a weighted-chance strategy: from the same critical thinking level (50%), next level (30%), or next aspect (20%). When it runs out the questions of all aspects, it will circle back with a random question about the current (sub)section. This strategy can increase the coverage of possible question alternatives and could simulate how people may ask questions [8]. When users scroll across pages, CReBot will proactively switch to an unseen section of the new page and ask questions as such high proactivity could enable CReBot to provide sufficient facilitation to users [61].

3.3. Experiment

To investigate CReBot’s usefulness for facilitating routine paper readers in critical reading and its user experience compared with that of the static guidelines, we conduct a mixed-methods, within-subjects study with 24 participants (P1-24; 9 Females, 15 Males; age range 19 – 61, $M = 25.5$, $SD = 7.9$). We recruit them via social networks, word of mouth, and snowballing, and they have proven certificates of fluent English skills or are native English speakers. Fourteen participants are postgraduate students, five are undergraduates, and the rest five are researchers in the industries. They all have a routine need to read scientific papers. Twenty of them major(ed) in Computer Science (CS), and the others major(ed) in Cognitive Science, Design, or Electronic Engineering. In general, our participants have moderate experiences in reading CS papers ($M = 5.0$, $SD = 1.5$) and HCI papers ($M = 4.6$, $SD = 1.6$; 1 - No experience at all, 7 - A lot of experience), and they are interested in HCI research ($M = 6.1$, $SD = 1.0$; 1 - No interest at all, 7 - A great deal of interest). On average, their self-reported skills in critical reading is 4.1 ($SD = 1.2$; 1/7 - Extremely incompetent / competent).

3.3.1. Baseline: The Guideline Tool

We denote the baseline tool as Guideline (Fig. 2b), which takes the popular form of existing critical reading assistance (e.g., [4, 6]) with a documented list of questions that can be asked about the paper. We embed it in the left sidebar of the web app to minimize unfair comparison with CReBot due to other usability issues irrelevant to critical reading. Users can check the Guideline conveniently without switching to another webpage. A click on a title or subtitle will unfold/fold its corresponding content.

3.3.2. Task

Participants read two CHI late-breaking works (LBW; six pages excluding reference) critically with either the CReBot or Guideline as if they were going to share the papers in an HCI group meeting, and the audience would ask them questions. LBW “represents work that has not reached a level of completion or maturity that would warrant the full refereed selection process” and “should have potential to, with time, make a contribution to the body of HCI knowledge” [62] Paper 1 is about user perception with strategies of artificial intelligence in offering suggestions (3151 words), and paper 2 is about the importance of visualizing the human body in a virtual reality application (3279 words). Both papers contain a small amount of technical content

and are perceived with moderate difficulties to read in the post-survey (item: “The paper itself is difficult to read”; 1/7 - Strongly disagree/agree; paper 1: $M = 4.3$; paper 2: $M = 3.5$; $p > .05$, repeated measured ANOVA). None of the participants had read these papers prior to our study. We counterbalance the order of two tools/papers conditions using Latin Square to minimize the potential order effect. To reduce learning effects and minimize fatigue, we schedule the two reading tasks of each participant at least 24 hours apart.

3.3.3. Measurements

RQ1. i) Reading behaviors. We log the completion time of each reading task, the number of highlights and comments, and the number of times scrolling into the previous/next page. **ii) Perceived engagement and difficulty in the critical reading process.** As previous works suggest that learning with a bot agent could be more engaging than with a non-bot tool [12], we hypothesize that (**H1A**) participants in the CReBot condition perceive significantly more engaged in the process of critical reading than those in the Guideline condition. To assess the effect, we adapt three 7-point Likert scale questions (e.g., “I was absorbed in this critical reading process”; 1/7 - Strongly disagree/agree; Cronbach’s $\alpha = 0.852$) about focused attention from the perceived engagement scale [63]³. We also measure participants’ perceived difficulty in reading the assigned paper critically (one item adapted from [64]) to estimate how CReBot/Guideline affects their cognitive load. **iii) Perceived performance in the critical reading outcome.** As the responses to the critical thinking questions are open-ended and hard to be judged objectively, we instead obtain the participants’ 7-point Likert ratings of their perceived critical reading performance to understand how CReBot might help from the users’ point of view. We measure it from three aspects regarding deep examination of the arguments, identifications of possible flaws, and reinterpretation of the paper for improved clarity [65] (one item for each aspect, e.g., “I had a *deep examination* of some claims, their supporting points and/or possible counterarguments in this paper”). Since educational bots could improve learning gains compared to non-bot applications [12, 13], we hypothesize that (**H1B**) compared to Guideline, CReBot can significantly improve users’ perceived critical reading performance.

³Detailed measured items in the experiments are included in the supplementary materials.

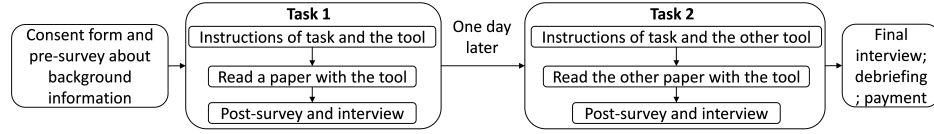


Figure 3: Procedure of the within-subjects (tool: CReBot vs. Guideline) experiment. In each task, participants read a given paper with either CReBot or Guideline based on their assigned order.

RQ2. i) Interaction with CReBot/Guideline. To understand how participants use CReBot, we log the number of their textual responses to the prompted questions, the number of questions added by users, as well as the number of times users click “Next question” and switch a (sub)section to be prompted questions about. For the usage of Guideline, we record the amount of clicking behaviors on the headings of each section category. We also ask for their frequency of using the tool in the questionnaire and inquire under what circumstance they use it in the interview after each task. **ii) Interruption and technology acceptance.** To evaluate user perceptions towards CReBot/Guideline, we first ask participants to rate the levels of reading interruptions by the tools (adapted from [61]). Then we adapt the technology acceptance model [57, 66] to measure the following in each condition: *usefulness* (four items, e.g., “I find the CReBot useful in my critical reading process”; Cronbach’s $\alpha = 0.890$); *easy to use* (four items, e.g., “I find it easy to get what I want from CReBot”; $\alpha = 0.841$); and *intention to use* (two items, e.g., “I intend to be a heavy user of the CReBot when I want to have a critical reading on the papers”; $\alpha = 0.862$). We average the ratings of multiple questions as the final score for each aspect in the acceptance model. Considering that CReBot can interactively prompt questions like a human peer, we hypothesize that in comparison to Guideline, users perceive CReBot to be **(H2A)** significantly more useful, **(H2B)** easier to use, and **(H2C)** of a stronger intention to be used in the future.

3.3.4. Procedure

Fig. 3 illustrates the study procedure conducted online via Zoom. At the beginning of each task, we first present task instructions and demonstrate the web app interface using a third example paper via a shared screen on a demo website. Then participants proceed to the main task and start reading the given paper with CReBot/Guideline. We allocate 35 minutes for each

reading session based on a pilot study with two users and tell the participants that they can finish early or take more time if needed. We also inform them that we do not restrict whether, when, and how they use the tool. After reading each paper, participants rate their engagement in the reading process, perceived performance, and perceptions about the tool in a post-survey. We further conduct a semi-structured interview with them to make sense of the ratings and collect feedback on when they use the tool. Upon completion of two tasks, we ask which tool they prefer and why as well as suggestions to improve it. After debriefing, each participant receives USD \$20 as compensation.

3.4. Analysis and Results

We first check whether the order of tool/paper conditions and the tool-paper assignment affects our quantitative ratings via a set of mixed ANOVA tests (tool/paper order or tool-paper assignment as between-subjects, tool as within-subjects) on each rating. Neither the main effect of the tool/paper order or tool-paper assignment nor its interaction effects with the tool are significant. We also try to include participants' backgrounds (e.g., experience in reading HCI papers) as covariates of the ANOVA model and find its effect to be insignificant. Hence, in the following repeated measured ANOVA analysis, we treat the tool (CReBot vs. Guideline) as the only within-subjects variable. For each ANOVA, the assumption of equal variance holds according to Macuchly's test of sphericity [67]. For the interview recordings, two of the authors transcribe them into text and conduct a thematic analysis [68] subsequently. They first familiarize themselves by reviewing all the interview data independently, and after discussion, they form a list of codes. After several rounds of coding with comparison and discussion, they consolidate different codes into potential overarching themes, which are CReBot/Guideline's pros and cons (Table 1), usage patterns of the tools, and suggestions for improvement. We count the occurrences of codes and incorporate these qualitative findings in the following presentation of our results.

3.4.1. RQ1 Results

Table 2 summarizes the RQ1 results about CReBot's impact on user's critical reading process and outcome in comparison to the Guideline. **i) Reading behaviors.** In general, participants spend significantly more time on reading a paper with CReBot than they do with the Guideline ($F(1, 23) = 16.83, p < .001, \eta^2 = .42$). They actively read papers in their tasks, with no

Table 1: Summary of routine paper readers’ interview responses to pros and cons of CReBot/Guideline in Stage 1.

	Pros (# of participants mentioned)	Cons (#)
CreBot	Conversational style of interaction (10); Checklist for examining papers deeply (8); Help understand the paper’s idea (6); Contextual questions (10); Auto question update + selection panel (4); “Export all chat log” feature (10)	Bring distraction (2); Some questions not matched to the reading context enough (7); Malfunctions of auto question update (9)
Guideline	Checklist for examining papers deeply (8); Systematic and detailed tutorial for critical reading (10); More freedom for exploration (3)	Bring distraction (4); lack of interactive features (8); Too much content on a page (16)

significant differences ($p > .05$) between the two conditions regarding the number of highlights, comments, and scrolling across pages (Table 2i). However, we observe more notes about users’ thoughts in the CReBot condition as they additionally respond to in total 129 what, 33 how, 13 why, and 85 how well questions from the bot in the chat window. This indicates that participants exercise critical thinking to understand and judge the papers more often with CReBot.

ii) Perceived engagement and difficulty in critical reading. In general, the participants feel that they are significantly more engaged with the reading material in the CReBot condition than in the Guideline condition; $F(1, 23) = 19.16, p < .001, \eta^2 = .45$; **H1A** accepted. Ten participants explicitly mention that CReBot’s conversational style of interaction (Table 1) helps them focus on critical reading. *“I like the bot. It asks me questions like my advisors to keep me thinking during the reading process”* (P24, Male, age: 22). No significant difference is found regarding the perceived difficulty in reading the paper critically with both tools ($p > .05$).

iii) Perceived performance in critical reading outcome. On average, the participants feel that they do equally well in both conditions in terms of having a deep examination of the arguments as well as identifying possible flaws in the authors’ reasoning ($p > .05$). They comment that both CReBot (eight users) and Guideline (also eight users) can serve as a checklist for examining the papers more deeply. *“After I took a pass on the paper, I chatted with CReBot. It asked me about the strengths and weaknesses of this paper, which reminded me to examine the paper’s weak and novel points”* (P21, M, 23). *“If I read the paper by myself, I would have blind spots. The Guideline reminded me to think of my goals for reading this paper and how it can contribute to my future research. I checked the discussion part and*

Table 2: **RQ1** results in the stage 1’s experiment with 24 routine paper readers. 7-point Likert scale (1/7 - Strongly disagree/agree) for ii) and iii). Mean (SD). ** : $p < .01$, *** : $p < .001$. #: number of occurrences.

	i) Reading behaviors				ii) Critical reading process	
	Time (min) ***	Highlight #	Comment #	Scroll pages #	Engagement **	Difficulty
CReBot	36.4 (8.4)	11.8 (8.2)	2.4 (3.6)	58.0 (33.6)	5.49 (1.07)	3.42 (1.38)
Guideline	30.9 (8.4)	13.3 (10.3)	3.5 (4.7)	58.0 (33.2)	4.54 (1.34)	3.38 (1.24)
	iii) Perceived performance					
	Examine arguments		Identify flaws		Reinterpret the paper **	
CReBot	5.17 (1.27)		4.63 (1.41)		5.08 (1.41)	
Guideline	4.79 (1.06)		4.88 (1.26)		4.13 (1.26)	

Table 3: **RQ2** results. Note that one participant may use the tools at multiple timings (*before/during/after* reading the full paper or each *section*). 7-point Likert scale (1/7 - Strongly disagree/agree) for subjective measures. Mean (SD). ** : $p < .01$, *** : $p < .001$.

i) Interaction with CReBot/Guideline							
	Frequency of usage * * *	Timing of usage #					
		Before read	During reading the paper			After read	Rare usage
			Bef sec	In sec	Aft sec		
CReBot	5.67 (1.46)	3	9	6	10	2	0
Guideline	3.13 (1.42)	9	2	7	1	5	4
ii) Perception towards CReBot/Guideline							
	Interrupt reading	Acceptance					
		Usefulness * * *		Easy to use **		Intention to use * * *	
CReBot	3.00 (1.50)	5.77 (0.71)		5.58 (0.79)		5.92 (0.99)	
Guideline	3.04 (1.88)	4.49 (1.07)		4.64 (1.40)		4.52 (1.65)	

made highlights accordingly” (P10, Female, age: 27). However, participants find that they do better in reinterpreting the paper for improved clarity under CReBot’s assistance ($F(1, 23) = 8.05, p < .01, \eta^2 = .26$). **H1B** is thus partially accepted. By answering CReBot’s questions, users can understand the paper’s ideas better, as suggested by six participants. “*CReBot helped me sort out the pipeline of the paper in my own way*” (P8, M, 22).

3.4.2. RQ2 Results

i) Interaction with CReBot/Guideline. As shown in Table 3i, overall, participants rate that they refer to CReBot for critical reading assistance significantly more frequently than to Guideline in the sidebar; $p < .001, F(1, 23) = 56.06, \eta^2 = .70$. In the CReBot condition, participants frequently click the “Next question” button ($M = 17.79, SD = 16.74$) and

write down their critical thoughts in response to a question in the text box ($M = 11.04, SD = 8.35$). Four participants add their own critical questions and consider it a practical function to record their thoughts. *“I can add questions to CReBot, which is similar to taking notes of my personal critical thoughts about the paper”* (P21, M, 23). On average, participants switch to a different (sub)section on the current page 2.42($SD = 2.22$) times and select a specific part of the paper from the dropdown menu 3.21($SD = 3.64$) times. When using Guideline, they click the headings in the outline to unfold the content of interests for around 11.96($SD = 6.84$) times.

Regarding the **timings** for using the tools, our thematic analysis on participants’ interview responses to “when did you usually use the tool” reveals quite a difference between two conditions (Table 3i). Participants commonly use CReBot as they are reading the paper while using Guideline before they start to read the paper content or after they go through the full paper. They interact with CReBot either right before digging into each section (nine users) so as to *“bring the questions to start reading”* (P5, M, 27), in the middle of reading a section (six) in cases of encountering *“some difficulties in understanding the paper in depth”* (P6, F, 23), or after going over a section (ten) to *“check if they have processed and understood this part thoroughly”* (P15, F, 25). This shows that CReBot encourages users to exercise critical thinking as they read each part of the paper.

ii) Interruption and technology acceptance. Table 3ii shows the participants’ ratings on their perceptions of CReBot and Guideline. Both tools generally make little **interruption** to the users’ reading processes. Yet, a few participants report that CReBot (two users) and the Guideline (four) in the sidebar may occasionally distract them from their reading materials. *“The interaction with CReBot shifted my attention away from the paper”* (P19, M, 21). In terms of **technological acceptance**, participants rate CReBot to be significantly more *useful* for critical reading than the Guideline in the sidebar; $F(1, 23) = 45.50, p < .001, \eta^2 = .66$; **H2A** accepted. While several participants acknowledge both tools’ usefulness as a checklist for in-depth paper examination (Table 1), ten participants particularly value the contextualized questions prompted by CReBot. *“CReBot’s questions were connected to my pace of reading and really helped me digest the corresponding section”* (P16, F, 23). However, seven people using CReBot find that *“some questions are still not well customized to the given paper”* (P12, M, 23). For the Guideline, the lack of interaction features prevents it from realizing its full value, as suggested by eight participants. *“The Guideline is not useful*

as *I cannot write down my thoughts to the questions*” (P21, M, 23). Yet, 12 users consider the Guideline as a “systematic” and “detailed” tutorial for critical reading. *“The Guideline looked like a manual which provides a detailed approach for reading papers critically”* (P20, M, 25).

Apart from the usefulness, participants feel that CReBot is significantly **easier to use** than Guideline (Table 3b); $F(1, 23) = 12.17, p < .01, \eta^2 = .35$; **H2B** accepted. Four participants comment that CReBot’s feature of updating questions automatically when they scroll onto a new page is “convenient” (P15, F, 25), and the panel for switching/selecting (sub)sections to be asked questions about is “flexible” (P18, F, 25). Nevertheless, nine users report a glitch in the current design of this question updating feature when CReBot fails to recognize their intention behind the scrolling action, e.g., *“when I scrolled back and forth to find answers for its question, the bot popped up another one”* (P24, M, 22). For the Guideline, while three users feel that it provides *“more freedom for exploration”* (P4, M, 30), 16 participants complain that it conveys too much content at the same time. *“It is too wordy. Sometimes I want to find the needed questions, but I have to go through all questions of that part to get those key points”* (P15, F, 25). All in all, participants have a significantly stronger **intention to use** CReBot than the Guideline in their future critical reading practices; $F(1, 23) = 25.02, p < .001, \eta^2 = .52$; **H2C** accepted.

In addition to the strengths of CReBot mentioned above, ten users particularly favor its “Export all chat log” feature, saying that it *“really helps me if I need to present this paper by reminding my critical thoughts about the paper”* (P13, M, 61). Besides, 14 participants are strongly positive that CReBot could improve their critical reading skills after a long-time usage. *“CReBot is useful. If I use it for several weeks, I believe that I can better grasp and examine the key points of the papers and form a consistent critical reading habit.”* (P23, F, 23). Moreover, readers may gradually reduce the use of both tools in the long run, as anticipated by five participants. *“Regardless of which tool [is employed], users will construct their own critical reading patterns after using it multiple times, and they will rely less and less on it”* (P17, M, 27). This is actually one of the goals of CReBot and matches the theory of scaffolding [27], which states that as learners become more independent in their thinking, the support from others can gradually fade away. For the interaction design, eight participants suggest that CReBot could be more dynamically integrated into the reading interface. For example, its chat window can *“pop up when users make a highlight”* (P20, M, 25) or can be

“attached to related paragraphs when asking questions” (P24, M, 22).

In summary, for users who often read research publications and have general research experiences, CReBot with interactive question prompts can significantly better engage them in the critical reading process and enhance their perceived performance in paper reinterpretation compared to the Guideline. Participants frequently interact with CReBot as they go over each section. They consider it significantly more useful and easier to use, and they favor its chat-like interaction that maintains their attention and its contextualized questions that stimulate thinking. Four participants mention that both tools could be particularly useful for novices who are new to paper reading by *“teaching them how to read a paper critically and efficiently”* (P11, M, 22). This motivates us to further investigate whether and how the interactive support from CReBot would help novices read scientific papers critically in Stage 2, as there is an essential need to teach critical thinking skills to students [1] through the exercise of paper reading [69].

4. Stage 2: Customizing CReBot for Novices

Encouraged by the findings in Stage 1 and the need to teach critical thinking skills to students [1, 70], we proceed to our second-stage exploration on CReBot’s usefulness and user experience for novices of scientific research. The representatives are junior undergraduates who have little or no paper reading experience but have desires to learn critical thinking and interests in going for advanced academic studies. To begin with, we verify the needs and identify requirements of CReBot specifically for novices. In this section, we first present a speed dating study for this purpose and then describe our refinement on CReBot as well as the user experiment and results.

4.1. Design Requirements of CReBot for Novices

To explore novices’ potential needs and requirements of CReBot, we follow a user-centered design approach “speed dating” developed by Zimmerman and Forlizzi [17]. This approach can reveal user needs or desires for new products or services with a small sample of participants and has been used by design teams either online or offline [18, 17]. In our speed dating sessions, we seek to investigate the concepts of a bot with interactive questions and guidance to support novices in critical paper reading. Specifically, we explore their needs for associated question prompts when digging into each section

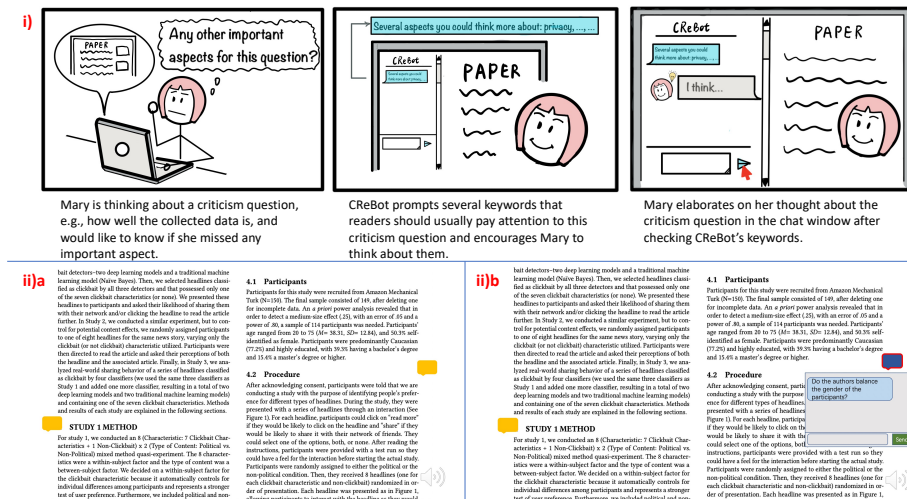


Figure 4: Speed dating materials used in Stage 2. i) An example storyboard used for examining novices' need for criticism question prompts from CReBot. ii) An example slide animation for user enactment in the design case of CReBot in a pop-up window – slide *a* is first presented, and then slide *b* is automatically played after a pre-set timing.

(step 2 in our compiled guidelines) as the results of our Stage 1's experiment imply that this step took most of the time for a critical reader. For a higher-level exploration of the CReBot's concept, we group the questions from four critical thinking levels to two sets based on their goals. **Comprehension questions** ask “what” is the purpose, claim, or premise in the text (e.g., motivation of the paper) – people usually can find direct answers to these questions from the paper. **Criticism questions** ask “how”, “why”, and “how well” (e.g., how well does the paper link to the related work) – readers usually need to analyze and evaluate the text to compose their own answers to these questions [71, 72]. We recruit eight undergraduate students (four females, four males, P1-8) from a local university with English as the teaching language. They are 18-20 years old ($M = 19.00, SD = 0.76$) and major in either computer science (CS), physic, math, or life science. They report limited experience in reading papers ($M = 3.38, SD = 0.92$; 1 - no experience at all, 7 - a large amount of experience) and almost no experience in reading HCI papers ($M = 1.25, SD = 0.46$) which are the reading materials in the speed dating study.

4.1.1. Speed Dating Materials and Procedure

Following the common practice of speed dating [17], we prepare 1) storyboards to verify the user needs and the usefulness of the potential CReBot's functions and 2) user enactment (UE) environment to explore novices' preferred interaction designs of CReBot.

Storyboards. We present the potential usage scenarios of CReBot to participants via storyboards with abstract bot components rather than the workable version in the first stage to keep the exploration more open. Participants view five storyboards one by one in a fixed order that explore the following needs brainstormed by our design team: comprehension questions, response to readers' answers to the comprehension questions, hints on where to find possible answers to comprehension questions, criticism questions, and hints to think of the criticism questions. Each storyboard consists of three figures with associated text explanation about a reader's problem during paper reading, the CReBot's assistance, and the reader's reaction to the bot (Fig. 4i) ⁴.

User enactment. Results of Stage 1's study suggest that users may use CReBot at different timing and favor different forms of integration into paper content. We thus explore through user enactment novices' preferred interaction design in 2 (location: sidebar vs. pop-up window on paper content) $\times$ 3 (timing: before vs. during vs. after reading related parts) conditions illustrated as separate animated slides (Fig. 4ii). In each condition, participants first see a full-screen slide that depicts the reading interface before the CReBot taking any action. After a given timing (i.e., 2, 24, 46 seconds ⁵), the animation depicting the assigned bot action, i.e., asking a question in the sidebar or in the pop-up window, will be automatically played. We randomize the order of the five slide animations and prepare each animation with a different paragraph from CHI papers to read and a CReBot's question to ask. We conduct the speed dating online via Zoom with participants' consent and start by the five storyboard tasks, in which they are asked to 1) empathize with the given persona; 2) report the frequency of facing a similar problem and the perceived usefulness of the bot's support in a 7-point Likert scale; and 3) share how they usually deal with the problem and what

⁴The storyboards and enactment slides are attached in the supplementary materials.

⁵The time is estimated based on the average time that two authors spend in reading each given paper paragraph.

other functions they need for the bot. Next, participants enact with CReBot via six slide animations in a random order, during which they are required to 1) read the assigned paragraph with the bot; 2) indicate likeability (1 - 7 points) of the bot's interaction design and why; and 3) explain how they would answer the bot's question. Each participant spends around 40 minutes for the whole study and gets \$6.2 as a token of appreciation.

4.1.2. Findings and Derived Design Requirements

The user needs for CReBot that asks comprehension and criticism questions and gives corresponding guidance are confirmed via the storyboards, as indicated by the participant ratings across the five scenarios: 1) frequency ($M = 4.00 - 5.25$; 1 / 7 - Never / Always) of various problems encountered in paper reading and 2) usefulness of CReBot ($M = 4.88 - 5.63$; 1 / 7 - Not useful at all / Very useful) for problem mitigation⁶. Participants comment that they normally get questions and hints from “*professors or online forums*” (P1) or “*senior students*” (P5, 6, 7) when facing issues depicted in the storyboards, suggesting that the CReBot could be a “*natural alternative for offering similar support*” (P2). For the interaction design of CReBot, results from user enactment show that participants generally prefer the bot in the sidebar ($M = 5.17$) rather than in the pop-up window ($M = 4.46$; 7 - like it very much); repeated measures ANOVA, $F(1, 14) = 2.48, p = .11, \eta^2 = .15$. The pop-up design has lower ratings as it would “*unexpectedly occlude the paper content*” (P1, 3, 4, 5). On average, participants express higher likeability to the CReBot when it proactively asks questions ‘after’ ($M = 5.38$) users finish reading related parts, compared to the ‘before’ ($M = 4.69$; Bonferroni post-hoc test $p = .54$) and ‘during’ ($M = 4.38$; $p < .05$) timings; repeated measures ANOVA, $F(2, 6) = 6.22, p < .05, \eta^2 = .67$. However, it is very difficult to track people's reading focus on the computer screen without access to webcams or additional eye-tracking devices [73], and proactive intervention at an unexpected timing would “*be distracting*” (P5, 7), similar to the negative feedback for CReBot's proactive questions in the first-stage study (Table 1). Instead, all participants suggest that CReBot should offer options for users to access its support at any time they want. They also expect various means to answer the bot's questions, including typing (mentioned by 8 participants) or copying & pasting (6) in the message box, as well as

⁶The ratings for each storyboard are in supplementary materials.

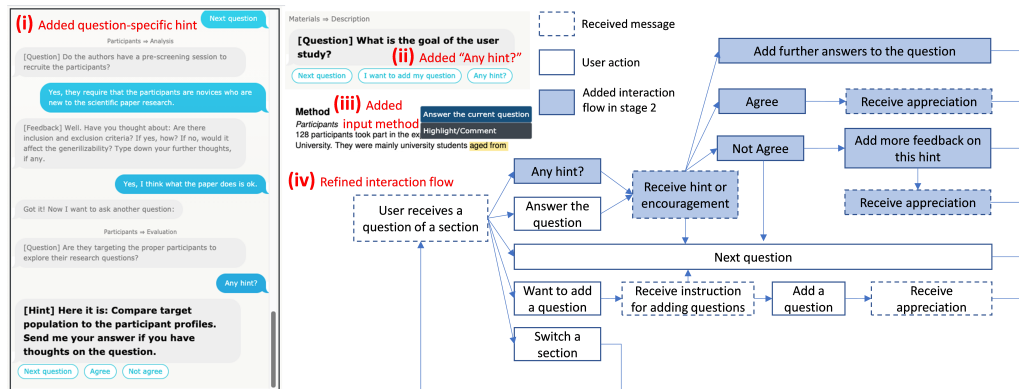


Figure 5: (a) Refined CReBot for assisting novices in critical paper reading in stage 2. Compared to previous version in stage 1, it mainly adds specific hints/feedback (similar content but triggered differently, inspired by the scaffolding interaction logic of Sara [13] and QuizBot [12]) that encourage thinking of each question (i, ii), an “Answer the current question” option for selecting the text from the paper as an answer (iii), and interaction flow for the question-specific hints (iv).

highlighting the corresponding paper content as a quick answer (6). Based on these findings, we derive two design requirements for further customizing CReBot to engage novices in critical paper reading:

DR1: CReBot should ask novices with comprehension and criticism questions and provide detailed hints to each question, so as to offer sufficient support for critical thinking.

DR2: CReBot should stay in the sidebar, offer multiple ways for user input, and proactively ask questions only when it can accurately track user’s reading pace; otherwise, it should let users initiate the interaction.

4.2. Refined CReBot for Assisting Novices

We refine CReBot for novices based on the aforementioned requirements. Compared with the previous version depicted in Fig. 2, the refined CReBot has the following changes (Fig. 5). *First*, it removes the high-level guidance and concentrates on providing question prompts to help novices dig into each paper section. This is to keep its functions focused on the most time-consuming step of critical reading and make them easy to understand for novices. *Second*, it provides teacher-like hints pre-compiled by HCI researchers for each critical thinking question (DR1). Correspondingly, CReBot extends its interaction flow to handle the user interaction with its

hints (Fig. 5i, ii, iv). For example, it offers options for users to agree or disagree with the hints, which can serve as a means to encourage further critical thinking and as user feedback for improving the CReBot. *Third*, it offers an additional input option for users to select the text of the paper and directly answer the current question as shown in Fig. 5iii (DR2). *Fourth*, it deducts the proactive question update feature when users scroll across pages as it could not accurately capture user’s reading pace and might cause disturbance (DR2). We leave this proactive feature to future work when there is a better reading pace inference solution with minimum sensing load.

To implement the second change, we invite three HCI researchers (males, age: 26, 25, 23) who got a Ph.D. degree or are Ph.D. students who majored in HCI to pre-compile question-specific hints in a brainstorming workshop. The first author, as one of the three participants, prepares the workshop materials, including common CHI section categories with titles and example critical thinking questions from the first-stage study (Section 3.1) as well as templates of hints (e.g., from [74, 75]). Inspired by the online critical thinking guidelines [72, 71], we ask brainstormers to contribute questions from three perspectives – “comprehension - description” (i.e., what), “criticism - analysis” (how, why), and “criticism - evaluation” (how well) (DR1) – which come to their mind when reading the paper contents under each section category. They also need to produce ideas about where one may look for possible answers or how to approach each proposed question. The online brainstorming session lasts for about two hours and results in a total of 103 question-hint pairs, which are further reviewed and refined by a professor who has published over 20 CHI papers. Fig. 5i shows some example pairs, and the full dataset is provided in the supplementary materials.

4.3. Experiment

Our evaluation of the refined CReBot for assisting novices in critical paper reading is guided by the following two research questions:

RQ3: Compared to the conventional Guideline that outlines the comprehension and criticism questions with corresponding hints, how would CReBot affect *novices*’ reading behaviors, perceived engagement and difficulty in the process, and perceived performance in the outcome of critical paper reading?

RQ4: How would *novices* use CReBot and Guideline in practice, and what are their acceptance towards such tools for critical paper reading?

[Participants] We conduct a within-subjects study with 20 participants (P1-20; 11 Females, 9 Males; age range 18 – 22, $M = 20.1$, $SD = 1.0$) re-

cruited via word of mouth in a local university with English as teaching language. Seven of them major in computer science (CS), and the rest major in life science, math, finance, law, etc. Different from those who already have general research experience and moderate experience in reading CS / HCI papers in stage 1, the targeted participants in stage 2 are novices with little or no experience in reading CS / HCI papers ($M = 2.4/1.5$; 1 - No experience at all, 7 - A lot of experience). The user study design is similar to that in stage 1 (Section 3.3). **[Baseline]** The baseline tool Guideline has a similar design as shown in Fig. 2b but replaces the 363 critical thinking questions with the 103 question-hint pairs used by the refined CReBot. **[Task]** Participants read two CHI LBWs the same as those used in stage 1. From post-surveys, the perceived reading difficulties of paper 1 and 2 are 3.85 and 3.75; $p > .05$, repeated measured ANOVA; item “The paper itself is difficult to read”, 1/7 - Strongly disagree/agree. **[Procedure]** We follow the same experimental procedure in stage 1 as depicted in Fig. 3. **[Measurements]** For RQ3, the measures include the logged reading behaviors, rated perceived engagement and difficulty in the critical reading process, and rated perceived performance in the critical reading outcome, using the same post-survey as that for RQ1 (see Section 3.3.3). As for RQ4, we also adopt the same post-survey as used for RQ2 to assess participants’ interaction behaviors with CReBot/Guideline, perceived interruption, and technology acceptance of each tool. We additionally log timestamped user interaction with the question-specific hints and the “Answer the current question” feature (Fig. 5iii) of the refined CReBot.

4.4. Analysis and Results

Note that in this experiment, we do not intend to have a direct comparison about CReBot’s usefulness for routine paper readers and novices; instead, we explore the refined CReBot’s interactive question prompts for novice compared to documented question list. We rerun all statistical tests used in stage 1 (Section 3.4). The main effect of tool/paper order, its interaction effects with the tool, and the effects of participants’ backgrounds (e.g., paper reading experience) on each quantitative rating are not significant. Therefore, we treat the tool (CReBot vs. Guideline) as the only within-subjects variable for the following repeated measured ANOVA analysis, with the confirmed assumption of equal variance based on Macuchly’s test of sphericity [67]. For the qualitative data from the transcribed interview recordings, we go through a similar thematic analysis as that in stage 1 and identify novices’ positive and negative feelings to CReBot/Guideline, their usage patterns of

Table 4: **RQ3** results in the stage 2's experiment with 20 novices. 7-point Likert scale (1/7 - Strongly disagree/agree) for ii) and iii). Mean (SD). * : $p < .05$, + : $.05 < p < .1$. #: number of occurrences.

	i) Reading behaviors				ii) Critical reading process	
	Time (min)	Highlight #	Comment #	Scroll pages #	Engagement	Difficulty **
CReBot	33.8 (5.8)	8.3 (7.5)	4.9 (3.9)	66.7 (51.6)	4.44 (0.88)	4.50 (1.32)
Guideline	32.9 (4.8)	9.7 (11.5)	3.1 (4.8)	62.5 (25.6)	4.58 (1.17)	3.90 (1.33)
	iii) Perceived performance					
	Examine arguments		Identify flaws		Reinterpret the paper +	
CReBot	4.45 (1.00)		4.20 (1.32)		4.35 (1.42)	
Guideline	4.55 (1.15)		4.45 (1.19)		3.80 (1.64)	

the tools, and suggestions for improvement. We incorporate them into the presentation of the results below.

4.4.1. RQ3 Results

Table 4 summarizes the RQ3 results about CReBot's impact on novices' critical reading process and outcome in comparison to the Guideline. **i) Reading behaviors.** All participants actively read and annotate (i.e., reading time, highlight, comment, scroll pages; all with $p > .05$) the papers with both tools (Table 4i). **ii) Perceived engagement and difficulty in critical reading.** In general, participants feel that they engage in the critical reading process equally well with both tools ($p > .05$). However, participants consider it significantly more difficult to read the assigned papers critically with CReBot than with Guideline; $F(1, 19) = 5.52, p < .05, \eta^2 = .23$. "*The bot somehow made me stressful and reduced my reading speed, making the critical reading a more difficult task for me*" (P10, M, 20). This suggests that CReBot's might create a more cognitively demanding reading environment than Guideline for novices. **iii) Perceived performance in critical reading outcome.** On average, participants in both conditions have similar perceived performance regarding argument examination and flaw identification ($p > .05$), though they tend to feel that they perform better with CReBot in paper reinterpretation ($.05 < p < .1$).

4.4.2. RQ4 Results

i) Interaction with CReBot/Guideline. As shown in Table 5i), participants rate that they use CReBot for critical thinking significantly more frequently than Guideline in the sidebar; $F(1, 19) = 5.31, p < .01, \eta^2 = .22$.

Table 5: **RQ4** results. Note that one participant may use the tools at multiple timings (*before/during/after* reading full paper or *section*). 7-point Likert scale (1/7 - Strongly disagree/agree) for subjective measures. Mean (SD). **: $p < .01$, + : $.05 < p < .1$.

i) Interaction with CReBot/Guideline							
	Frequency of usage **	Timing of usage #					
		Before read	During reading the paper			After read	Rare usage
			Bef sec	In sec	Aft sec		
CReBot	4.35 (1.46)	2	6	6	6	2	0
Guideline	3.30 (1.17)	3	1	1	11	7	1
ii) Perception towards CReBot/Guideline							
	Interrupt reading +	Acceptance					
		Usefulness		Easy to use		Intention to use	
CReBot	3.60 (1.76)	4.76 (1.38)		4.13 (1.21)		4.27 (1.52)	
Guideline	2.75 (1.37)	4.95 (1.36)		4.73 (1.39)		4.43 (1.40)	

Participants frequently check the “Next question” ($M = 17.15, SD = 13.10$) and answer the bot’s questions ($M = 7.20, SD = 6.29$). Half of the answers are inputted by selecting the paper content and clicking the pop-up “Answer the current question” ($M = 3.70, SD = 3.80$) – a new feature in the refined CReBot (Fig. 5iii). Our participants also ask for hints (i.e., click “Any hint?”) quite often ($M = 4.75, SD = 6.70$) and actively express their feelings to the received question-specific guidance. In total, they “agree” to the bot’s guidance 78 times and do “not agree” to it ten times, suggesting that they carefully check the hints and that the pre-compiled hints generally are of high quality. Only one participant added his own critical thinking question, and this observation is acceptable as novices may still learn to read papers critically, and it could be difficult to contribute new questions to CReBot in our lab study. For the usage of Guideline, participants click the headings in the outline to unfold the content of interests for around 10.60($SD = 5.64$) times. Regarding the **timings** at which users checked the questions from the two tools, we have an interesting finding (Table 5i). With the Guideline, novices usually turn to the questions after reading each paper section (number of occurrences: 11) or the entire paper (7), while with CReBot, some of them refer to the critical questions before (6) or during reading each section (6). This implies that CReBot encourages some novices to adjust their reading habits from “read first” to “think first” or “think in situ”. “*CReBot’s questions caught my attention as I read each paper section, and I started to think about them and highlighted potential answers from the section as I proceeded. This makes my critical thinking on this paper more efficient*” (P3, F, 19).

Table 6: Summary of novices' interview responses to pros and cons of CReBot/Guideline in Stage 2.

	Pros (# of participants mentioned)	Cons (#)
CReBot	Good critical thinking questions (8); Inspiring hints (8); Flexible interaction (7); Contextual questions (3)	Some questions not matched to the reading context enough (3); Some hints are general (3)
Guideline	Good critical thinking questions (6); Inspiring hints (8); Clear structure (4)	Lack of interactive features (4); Too much content on a page (7)

However, three participants comment that they are not used to this change in their reading habits in the experiment. *"Taking the questions in mind before or during reading each section is not my familiar reading style. It is like being challenged by the bot, and I do not like this feeling"* (P1, F, 20).

ii) Interruption and technology acceptance. As shown in Table 5ii), both tools generally make little *interruption* (e.g., both $M < 4.00$) to the users' reading processes, though CReBot tends to be more interrupting than Guideline; $F(1, 19) = 3.03, p = .098, \eta^2 = .14$. In terms of *technological acceptance*, participants deem CReBot and Guideline equally useful and easy to use, and they have similar levels of intention to use both tools in the future ($p > .05$ in all three aspects).

We further dig into the qualitative data to probe what aspects of CReBot/Guideline work or do not work for novices (Table 6) as well as their expectations for future interactive critical paper reading support. In general, both tools are valuable for novices by providing good critical thinking questions, as suggested by eight/six users with CReBot/Guideline. *"The bot asked me some insightful questions that help me grab key points of the paper to think of"* (P19, F, 21). Eight participants also appreciate the question-specific hints from both tools, saying that *"they are inspiring for locating answers or getting an angle to criticize the paper"* (P11, M, 22). Similar to the qualitative results (Table 1) in stage 1, novices favor the flexible interaction features (mentioned by 7 users) and contextual questions (3) of CReBot and the clear and systematic structure (4) of Guideline. *"I like the conversational style of interaction with CReBot, which is flexible and interesting."* (P4, F, 19). *"The Guideline interface is simpler and more concise for me"* (P15, F, 20). Nevertheless, the Guideline is criticized by seven users for that *"it provides too much content at the same time, making it hard to read and use"* (P6, F, 21). Six participants mention that some of the CReBot's questions and hints still do not match the section content very well, and they

desire “more questions and hints that associate with the specific details of this paper and even some follow-up questions to the previous user responses” (P19, F, 21). Regarding the envisioned design of a future interactive critical paper reading support tool, participants suggest some additional features: 1) certain questions could be presented with “yes/no” or “multiple choices” option buttons to enable quick answers (commented by 3 users) and 2) it could have a panel in which readers can store their interested questions and hints for reflection and future usage (3).

In summary, for novices who rarely read scientific papers before, the current design of CReBot with interactive questions and associated hints may not be a better solution than the static checklist of questions and guidance (Guideline) for assisting critical paper reading. The pre-compiled critical thinking questions and guidance presented by either tool are generally valuable for novices. While CReBot is still favored for its flexible interaction and contextual support, it could interfere novices’ traditional reading habits and make them uncomfortable. We discuss possible reasons why CReBot is not deemed more helpful for novices than Guideline as that for routine paper readers in the next section.

5. Discussion

5.1. Reflection of CReBot for Assisting Routine Paper Readers and Novices

From our first-stage findings, CReBot with interactive section-level question prompts is more useful for engaging routine paper readers in critical thinking than Guideline. These people can be postgraduates and researchers who have experience in scientific research and are already familiar with the scenarios (e.g., group meeting, rebuttal) of responding to questions and/or criticisms regarding the papers they read or write. CReBot simulates the active Q & A activity in these scenarios to help this group of users think critically before, during, and after reading each section of a paper (Table 3), which is proven beneficial in our stage 1 study. This is in line with the findings in [45] that the questions which users may ask for understanding artificial intelligent (AI) systems can guide design practitioners to think of and create user-centered explainable AI applications.

In the second-stage exploration, we refine CReBot with additional teacher-like hints that helps novices approach each critical thinking question. While the questions and hints are appreciated by the novices in the user study, the

interactive way of presenting them by CReBot does not outperform the traditional method that documents all information as a static list (Table 5ii). One possible reason is that CReBot's interactive questions lead some novices to think before or during reading each paper section (Table 5i), which might not be their familiar and comfortable ways of reading. Furthermore, as the novices in our study use CReBot more frequently than Guideline, they possibly face questions that are hard to answer more often due to the lack of prior experience in scientific research, which may make the critical reading task more difficult for them (Table 4ii). For example, the question "How do the authors interpret the numerical results statistically?" may be difficult for those novices who are unfamiliar with common statistical analysis methods used in (HCI) research [76]. Theoretically, this could be also explained by the Yerkes-Dodson law [77], which states that people's task performance increases with physiological or mental arousal, but only up to a point. The critical paper reading tasks would be more intellectually demanding for the beginners who lack prior research experience and thus they may require a lower level of arousal for optimal performance [78]. Compared to the static Guideline, the interactive questions from CReBot might cause higher arousal than the optimal level, leading to a potential decrease of the bot's usefulness and user experience for novices in critical paper reading.

Our two-stage studies imply the need to customize the interactive reading support tools for users of different research experiences. For routine paper readers, the reading support tools can imitate the human interaction in offline co-located group paper discussions [79], such as question prompts in our case, commenting, and association of related works. For novices, a more complete picture of critical reading knowledge space (e.g., the organized question list in Fig. 5b) rather than the fragment questions prompted by CReBot is needed [80]. The reading support tools can act as a teacher to present this type of structured knowledge space to novices via a scaffolding manner [81], e.g., raising questions, checking users' answers, and giving corrective feedback if applicable. To personalize the reading support based on users' levels of critical thinking skills, the interactive tools can invite the readers to ask questions about the papers like what our CReBot currently supports. These questions raised by readers can be further compared with the high-quality critical thinking questions to assess their competency in paper reading.

5.2. Extension of CReBot to Technology-Enhanced Learning Communities

For other paper domains. In the development process of CReBot, we create a question bank for paper reading. It includes 363 critical thinking questions from interviews with domain researchers in stage 1 (Section 3.1) and 103 question-hint pairs from a brainstorming workshop in stage 2 (Section 4.2). As a proof-of-concept, these questions are customized by HCI researchers and mapped to the common CHI paper sections in the scope of this work. Following the same pipeline, researchers could easily adapt this initial question pool to other research domains, enabling CReBot to provide critical reading assistance specific to the nature of these domains.

For critical writing and review. Given CReBot's benefits for routine paper readers, we anticipate that CReBot can also facilitate then in scientific writing and paper reviewing, which are two other common research activities that involve critical thinking. Critical writing skills lie in the capability to convince readers to accept the authors' claims [3] and are reciprocal to the critical reading [82]. Therefore, it is promising that authors can leverage our CReBot to self-check if their writings sufficiently address potential readers' concerns. Reviewers for conferences and journals can also employ our CReBot to identify and articulate the strengths and weaknesses of a submission, and provide constructive comments in terms of how the submission could be improved. Future research can explore the potential extensions of CReBot to these contexts. For example, to assist critical writing, designers can add features such as persuasiveness of the arguments [57] and recommended similar arguments from published papers [56, 83].

For collaborative reading. Beyond the usage for individual paper reading, CReBot can also involve a group of people to read collaboratively, as hoped by five participants in the first-stage experiment. "*CReBot can record all users' questions and answers so that people can check others' thoughts.*" (P4 in the first CReBot evaluation study, M, 30). This idea is similar to the shared highlights and notes of Amazon Kindle [84], the collaborative critical reading platform [10], and cooperative note-taking system [85], which require human peers to form a reading group. We suggest that CReBot can be integrated into online groups for academic paper reading (e.g., PeerLibrary [86], Fermat's Library [87], Reddit *r/MachineLearning* [88]). In this case, CReBot not only acts as a facilitator of critical reading but also a "social organizer" [89] who connects users with other community members based on their critical thoughts on the same paper and encourages them to share ideas

and hold discussions. At the same time, CReBot will be able to evolve itself iteratively with users' question-answer pairs.

5.3. Design Considerations for Interactive Critical Reading Support

5.3.1. Providing Proactive Assistance when Users Get Stuck

During the critical paper reading process, users especially those new to scientific research (i.e., novices) may get stuck in answering the critical thinking questions from time to time. To offer sufficient facilitation in this case, CReBot could add more proactive mechanisms [61] such as auto-highlights of keywords, sentences, or paragraphs that might be of interests to the readers (e.g., by tracking discourse markers like “we found that” [34]) when they get entangled in a question. Common ways of detecting such struggles to initiate proactive support include but are not limited to detecting extensively long pause at certain paper content through interface interaction logs or eye-tracking [90, 11]; the latter is more accurate but has the trade-offs of privacy concerns and possibly involving additional hardware.

5.3.2. Supporting User Customization of Prompted Questions and Guidance

As suggested by three participants in stage 2, CReBot can offer another panel in which users can easily store their interested questions and hints. For example, it can offer a “save” or “bookmark” checkbox next to every prompted question and guidance (Fig. 5i), and a check on this box can automatically copy this message to the designated panel so that users can easily reflect on their preferred critical thoughts to the questions later. This is like the “idea checking area” in MetaMap [91] that stores searched images for visual metaphor ideation and the “Saved Designs” view in the GRIDS [92] tool that lets designers record their design solutions.

5.3.3. Tutoring Novices via an Ad-Hoc Critical Thinking Quiz

Two novices in the second-stage experiment suggest that CReBot could better help them learn critical thinking if it can set up an ad-hoc quiz to test how well they have mastered the paper content and the critical reading skill. “For a given text, CReBot can ask us to identify its weak points and give us corrective feedback, such that we can learn how our critical thoughts can be improved” (P11 in the second CReBot evaluation study, M, 22). We thus suggest that with the known weak points in the paper, CReBot can offer such a quiz during novices' paper reading exercises. For example, we can leverage the comments of reviewers on some papers (e.g., those in the OpenReview

[93]) and extract the points that mention the papers' weaknesses (e.g., using keywords like "not convincing" and "need to improve"). CReBot can post a quiz about flaw identification when readers start to read related paper sections and provide corrective feedback with the pre-set answers. This is an instructional scaffolding strategy [81] commonly used in the classroom and recently in educational chatbots like Sara [13] and QuizBot [12] that are showed to be effective in helping students to learn.

5.4. Limitation and Future Work

Our work has several limitations. First, we show in stage 1 that routine paper readers find CReBot more useful for stimulating them to reinterpret the reading materials for improved clarity than the Guideline (Table 2iii). However, given that our focus is primarily on how users interact with the system, we do not carry out pre-post tests to measure changes in their critical understanding of the paper (e.g., the strengths and weaknesses of the paper) after using CReBot/Guideline. We thus do not get to evaluate CReBot's effectiveness objectively. Future work can derive metrics of critical thinking performance and examine how well CReBot can help users improve their mastery of scholarly publications. Second, our results are from a short-term study, in which participants read our selected CHI late-breaking works in a simulated scenario where they need to present the papers in a group meeting. In real-world contexts, people might read other styles of papers (e.g., CHI full papers with various topics and/or different types of contributions) and have different purposes for reading papers critically (e.g., homework for research seminar course and literature review [5, 3]). In the future, we need to explore the usage of CReBot in a long-term field study in which users can read papers of their interests to see if people's critical reading skill improves and eventually they could read papers critically without the bot. Third, we invite HCI researchers to contribute the section-level questions and guidance for CReBot, which are perceived helpful but sometimes do not match very well to specific papers as noted by the participants in two stages. Future work can explore automatic question generation based on the paper's content to provide more paper-specific questions and hints at scale. For example, it can use template-based methods that fill phrases (e.g., semantic role labels [94]) from the reading materials into some templates or use machine-learning models that are trained on labeled question-answer-context data [95] when such dataset is available in scientific paper domain.

6. Conclusion

This paper probes the design, usefulness, and user experience of the interactive question prompts for critical paper reading support. Our first-stage study aims to offer such facilitation to routine paper readers who have prior research experience and need to read papers critically. We build a CReBot system that interactively asks section-level questions from the pre-compile critical paper reading guidelines when users are reading each paper section. Our within-subjects experiment with 24 routine paper readers shows that CReBot can better engage them in the critical reading process and is perceived significantly more useful and easier to use than the static checklist-like Guideline. In our second-stage study, we further customize CReBot for novices who are new to scientific research by adding more question-specific hints, based on the identified needs and requirements in a speed dating study with novices. However, the results from another within-subjects study with 20 novices imply that CReBot is not perceived as more useful and easier to use than the static Guideline. Novices appreciate the critical thinking questions and associated guidance from either tool but the CReBot sometimes unexpectedly interferes with their conventional reading habits. The main takeaways of our works include 1) the CReBot prototype to exploring interactive question prompts for critical paper reading support, 2) demonstrated value of CReBot for facilitating routine paper readers, 3) requirements of interactive critical reading support for novices and how they work with CReBot, and 4) design implications for interactive tools to enhance learning. Meanwhile, we contribute a question bank with guidance that encourages critical thinking of scientific research publications.

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